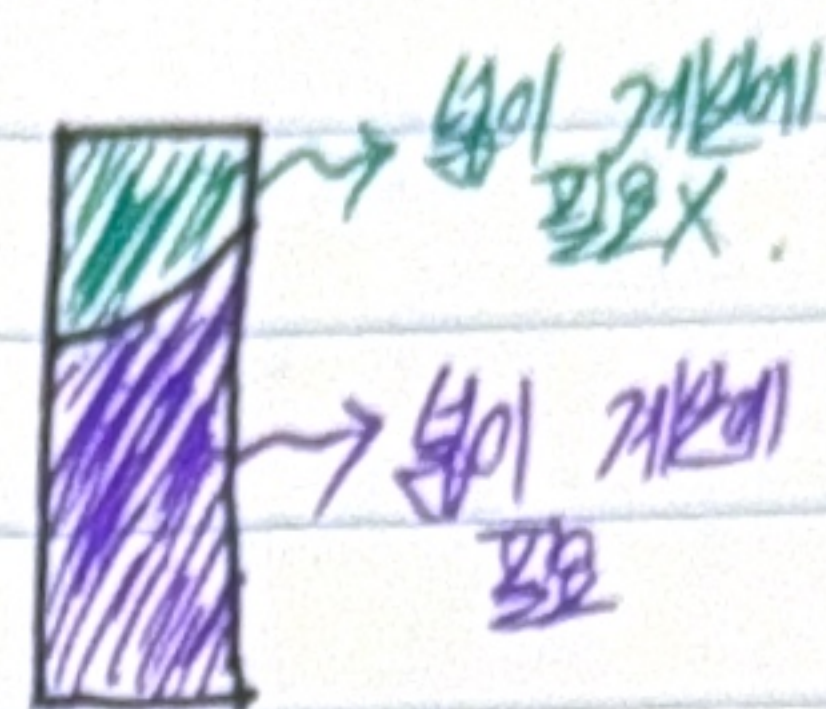
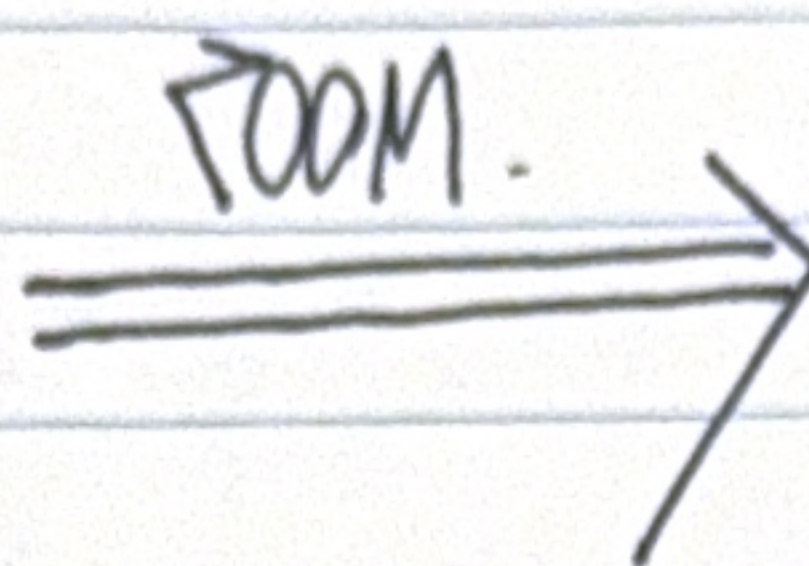
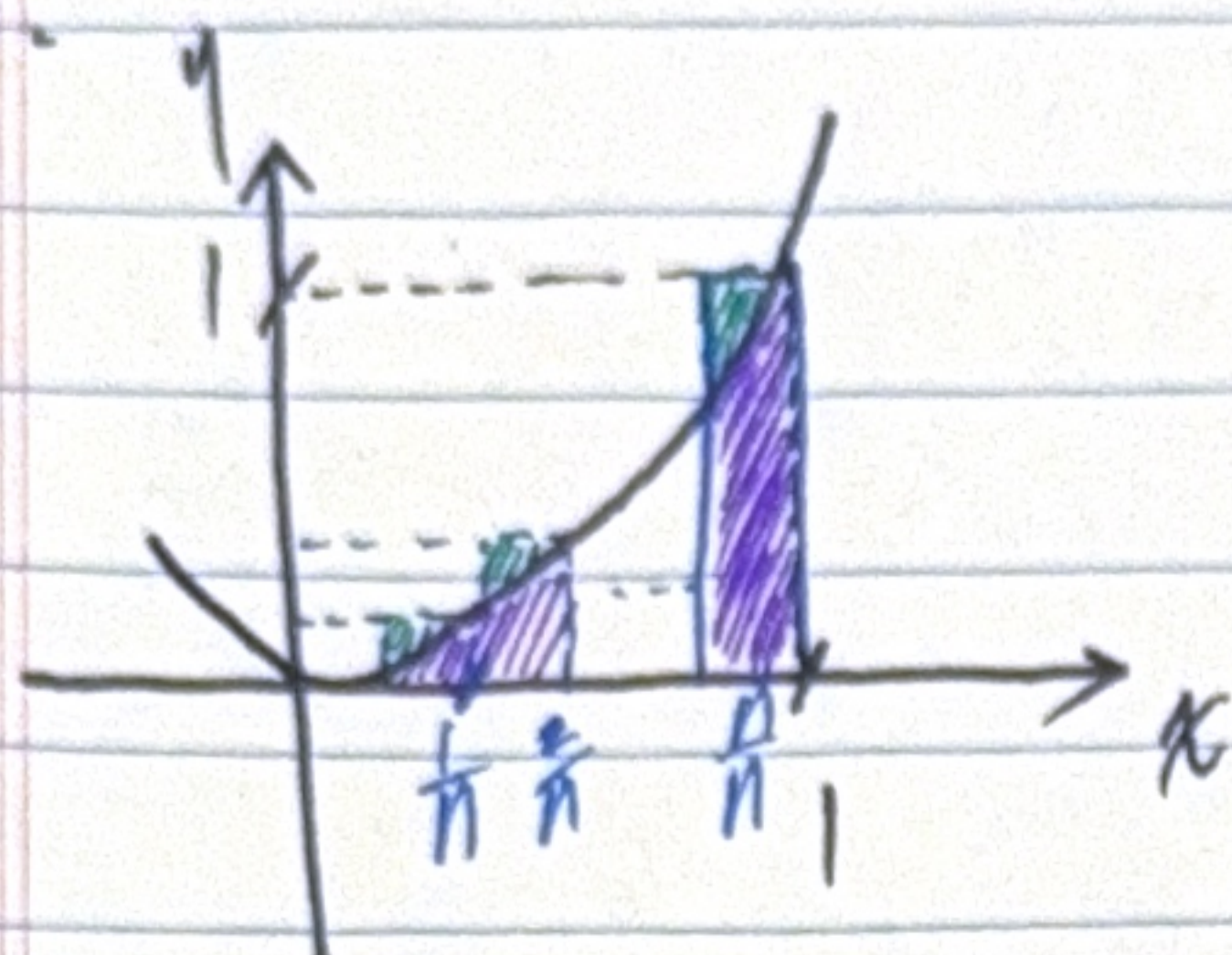


문제 3)

(1)



1)
$$S = \sum_{k=1}^n \left(\frac{k}{n} \right)^2 \cdot \frac{1}{n}$$

정확히 n. 1/n씩 k.

$$= \sum_{k=1}^n \frac{k^2}{n^3} \cdot \frac{1}{n}$$

$$= \sum_{k=1}^n \frac{k^2}{n^3} = \frac{1}{n^3} \sum_{k=1}^n k^2 = \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6}$$

1/n씩 k 1/n씩 k

$$= \frac{n(n+1)(2n+1)}{6n^3} = \frac{1}{3}$$

정확히 n.

2)

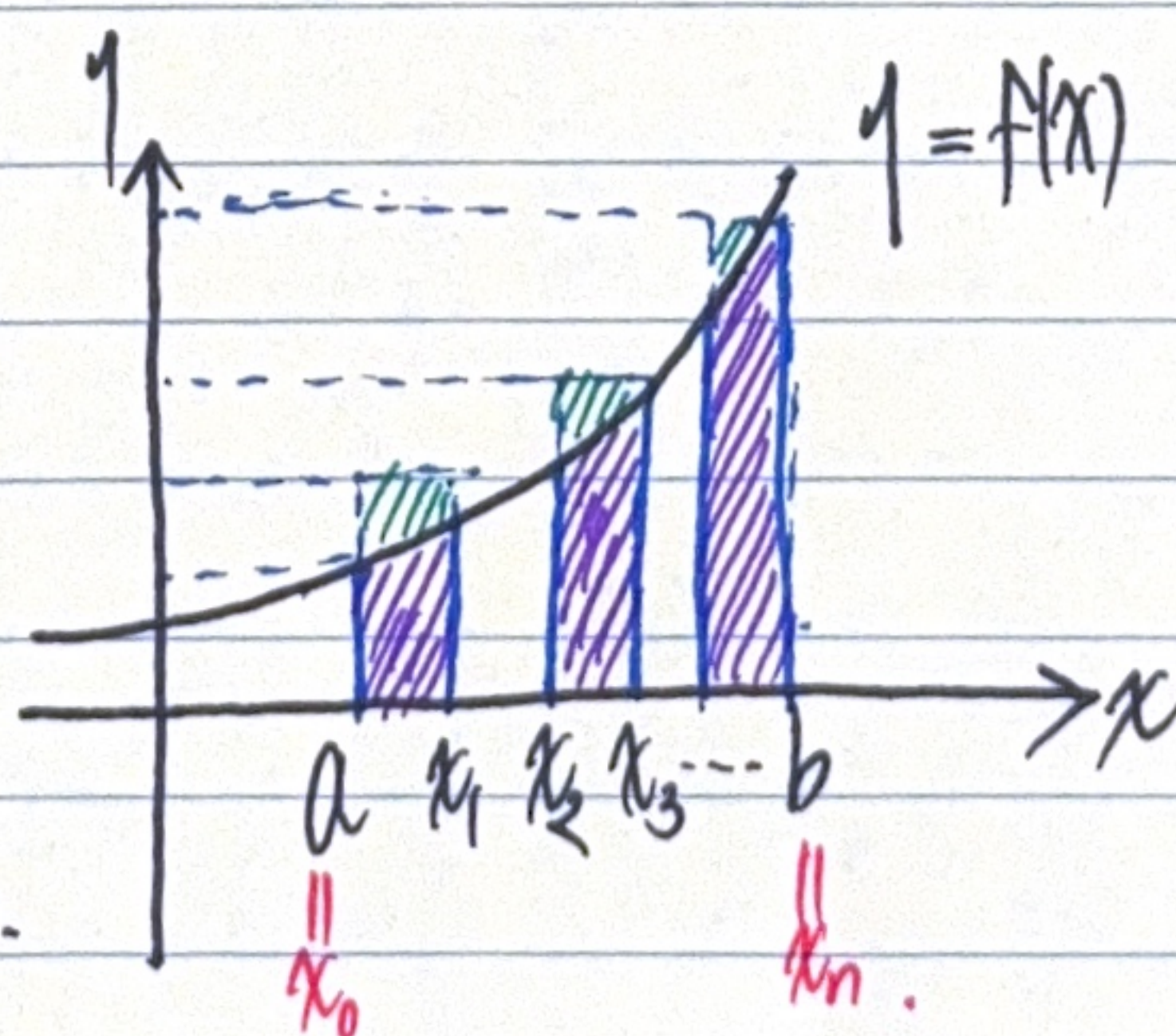
$$S = \sum_{k=1}^n \left[\left(\frac{1}{n} \right)^2 \frac{1}{n} + \left(\frac{2}{n} \right)^2 \frac{1}{n} + \left(\frac{3}{n} \right)^2 \frac{1}{n} + \dots + \left(\frac{n}{n} \right)^2 \frac{1}{n} \right]$$

$$\Rightarrow \int_0^1 x^2 dx = \frac{1}{3}$$

$$\Rightarrow S_n = \sum_{k=1}^n f(x_k) \cdot \frac{b-a}{n}$$

= x_n - x_0

$$\Rightarrow \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k) \cdot \frac{b-a}{n} = \int_a^b f(x) dx$$



GF) 그래프의 넓이를 구하는 방법